



Objectives:

1. Proceed with the experimental/simulation work and data collection.
2. Proceed with data analysis and interpretation.
3. Able to analyze and interpret the data and make convincing arguments on the data analysis and interpretation.

Student's Name/Signature:
NAVINNESHSAAN

Date: 15 May 2026

Supervisor's Name/Signature:

Date:



Summary of weekly activities

During this period, the primary focus was validating the initial synthetic dataset and transitioning into Phase 3. Upon analyzing the first batch of automated simulations, unrealistic Time Difference of Arrival (TDOA) values were identified. Extensive troubleshooting revealed that traditional gradient-based extraction logic (dP/dt) struggled with the natural wave attenuation and simulated Gaussian SCADA noise, occasionally triggering on solver initialization transients or fixed-seed static rather than the true Negative Pressure Wave. To resolve this and optimize the data for Machine Learning, the extraction methodology was entirely overhauled. A re-simulation of the 200 Class 0 and Class 1 scenarios was conducted to capture the complete, raw time-series pressure data (15,000 data points per sensor) directly into a MATLAB cell array. This ensures the dataset perfectly preserves the physical shape and macro-trend of the pressure curves without relying on rigid, error-prone threshold equations.

With the robust dataset successfully packaged, Phase 3 has officially commenced using MATLAB's Deep Learning Toolbox. The raw time-series data was formatted and split into training and testing sets. Currently, a Recurrent Neural Network (RNN) utilizing Long Short-Term Memory (LSTM) layers is being developed and trained. The network is designed to directly analyze the continuous wave patterns to classify normal versus leak operations and predict the exact leak location (X). The immediate next step is to conclude the training process and test the model against the holdout dataset to evaluate its prediction accuracy and overall performance.

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UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF ELECTRICAL ENGINEERING

WEEK 6 - WEEK 9

Summary of weekly activities

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NAVINNESHSAAN

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